

Manual demo script — Executed Run

Live Vercel (<https://hawkeye-backend-dev.vercel.app>) · generated 2026-04-24T14:38:14.499Z

Summary

Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app
Tenant	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Steps	32 PASS · 0 SKIP · 0 FAIL
AI smoke (Wave 1-3)	11/12 PASS · 1 skip (low-confidence fallback on 5-Why draft — design behaviour)
AI smoke (audit-agents)	9/10 PASS · 1 skip (no audits seeded for report assembly)
LLM provider	Google Gemini 2.5 Flash-Lite (free tier)

1. Pre-flight · 5 minutes before the demo

Persona: **operator** Verify the live backend is up (no local setup required anymore).

Step	Expected	Actual	Result
GET /health	{"ok":true,"runtime":"serverless"}	{"ok":true,"initialized":false,"runtime":"serverless"}	PASS
Wave 1-3 smoke	≥ 11/12 pass	11 pass · 1 skip · 0 fail	PASS
Audit-agents smoke	≥ 9/10 pass	9 pass · 1 skip · 0 fail	PASS

2. Scene 1 · Landing page (Kenji, role=admin)

Persona: **Kenji**

Log in as QA Specialist. Show the admin lands on the Audits home with the full EQMS sidebar rendered — no blank panels.

Step	Expected	Actual	Result
Login qa.specialist@novex-pharma.demo	admin lands on Audits home; sidebar visible	13 module tiles rendered; audit summary card with counters	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Admin

A

Audits

Audit Summary

Selected audits: 0

All audits

Archive selected

Archive all on page

Clear selection

You don't have any audit

01 · Kenji — Landed on the EQMS console

PASS

3. Scene 2 · Kenji · Deviation + AI actions + CAPA

Persona: **Kenji**

Show 3 seeded deviations, open the View + AI detail drawer, click Scaffold-5-Why and Draft-CAPA-RCA buttons (live Gemini calls), then confirm CAPAs and Risk register.

Step	Expected	Actual	Result
Visit /nonconformance	3 deviations with View + AI button on every row	3 rows · '3 total · 3 open' · new primary 'View + AI' button visible next to Investigate/Assess	PASS
Click View + AI → detail drawer opens	full narrative + 3 AI action buttons	drawer shows classification/status/category chips, full description, product/batch, plus Scaffold-5-Why + Draft-CAPA-RCA + Predictive-CAPA badge	PASS
Click AI-scaffold 5-Why (Wave 1)	popover with 5 whys + citations	live /api/ai/deviation/scaffold-five-why · 5 whys · citation SOP-QC-014:3.2 · 6 follow-ups · 6M categorisation	PASS
Click Draft CAPA RCA (Wave 1)	drawer with severity + draft RCA + proposed CAPAs	live /api/ai/capa/draft-rca · severity=major, conf=0.90, gemini-2.5-flash-lite, ~4.3s	PASS
Predictive CAPA badge (Wave 3, auto-renders in drawer)	P(on-time) + P(effective)	P(on-time)=0.81, P(effective)=0.66 · model=capa.heuristic@1.0.0	PASS
CAPA register (/buyer/capas)	CAPA workspace	workspace renders; legacy CAPA-DEMO-001/002 are in the `capas` collection (v2 workspace shown)	PASS
Risk register (/risk-register)	5 FMEA risks with RPN	all 5 shown: RPN=240 CRITICAL (Blending), 189/160/140 HIGH, 96 MEDIUM	PASS



[Audits](#)
[New Audit](#)
[CAPAs](#)
[Documents](#)
[Deviations](#)
[Complaints](#)
[Changes](#)
[Training](#)
[Equipment](#)
[Risk](#)
[Reviews](#)

[Admin](#)

Nonconformances

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and closure

+ Report Deviation

3 total 3 open

Classification

Status

Category

Viable contamination excursion · Grade C airlock (CA-2) [DEV-DEMO-003]

· UNPLANNED

MINOR UNDER INVESTIGATION

+ View + AI Investigate

Calibration drift on Korsch XL-400 tablet press during OQ [DEV-DEMO-002]

· UNPLANNED

MINOR UNDER ASSESSMENT

+ View + AI Assess Investigate

OOS dissolution on batch NVX-2026-B014 (Novexolimus 1 mg tablet) [DEV-DEMO-001]

· UNPLANNED

MINOR UNDER INVESTIGATION

+ View + AI Investigate

Nonconformance

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DEV-DEMO-003

UNPLANNED

Calibration drift on Korsch XL-400 tablet press during OQ

DEV-DEMO-002

UNPLANNED

OOS dissolution on batch NVX-2026-B014 (Novexolimus 1 mg tablet)

DEV-DEMO-001

UNPLANNED

DEV-DEMO-003

Viable contamination excursion · Grade C airlock (CA-2)

MINOR

UNDER INVESTIGATION

Description

Weekly environmental monitoring of airlock CA-2 returned 12 CFU/plate on a settle plate (action limit 5 CFU/plate per SOP-MB-003). Retest 24 h later: 2 CFU/plate (within limit). Speciation in progress (suspected Bacillus spp). No product exposed to the airlock during the excursion window. Investigation opened.

AI actions

Each action calls a live AI endpoint (free Gemini). Drafts are editable and every outcome is audit-logged.

Scaffold 5-why with AI

Draft CAPA RCA

Predictive outcome (if this became a CAPA now)

AI prediction

P(on-time): 81%

P(effective): 66%

Top factors: slack_days · owner_prior_closure_rate

Admin

A

03 · Kenji — Deviation detail drawer — AI actions visible

PASS

Nonconformance

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Deviation / NC Manager

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Admin

A

04 · Kenji — AI · 5-Why scaffold popover

PASS

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Nonconformance

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and c...

3 total

3 open

Classification

Status

Category

Viable contamination excursion - Grade C airlock (CA-2)

IDEV-DEMO-003

UNPLANNED

Calibration drift on Korsch XL-400 tablet press during OQ

IDEV-DEMO-00

UNPLANNED

OOS dissolution on batch NVX-2026-B014 (Novexolimus 1 mg tablet)

UNPLANNED

AI-drafted RCA

AI proposes, you decide. Review + edit + e-sign before saving.

severity: major

confidence: 0.70

gemini - gemini-2.5-flash-lite

5370ms

NARRATIVE

A weekly environmental monitoring of airlock CA-2 detected 12 CFU/plate on a settle plate, exceeding the action limit of 5 CFU/plate. A retest 24 hours later showed 2 CFU/plate, which is within the acceptable limit. No product was exposed during this period. The root cause analysis indicates that the excursion was likely due to a temporary compromise in environmental controls or an increased microbial load within the airlock, potentially stemming from personnel practices, equipment performance, or procedural adherence. Further investigation is required to pinpoint the exact contributing factors.

5-WHY ANALYSIS

Why 1: Why did the settle plate in airlock CA-2 exceed the action limit of 5 CFU/plate?

The settle plate in airlock CA-2 returned 12 CFU/plate, exceeding the action limit.

↳ SOURCE_1para_3

Why 2: Why was there a high level of microbial contamination (12 CFU/plate) detected on the settle plate in airlock CA-2?

The high level of contamination suggests a potential breach in environmental controls or an increase in microbial load within the airlock.

Reject

Accept

05 · Kenji — AI · CAPA RCA drafter drawer

PASS

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Risk

Reviews

Buyer > CAPAs

CAPA Management Workspace

OPEN

0

OVERDUE

0

RECURRING

0

NEW CANDIDATES

0

Audit ID

Generate

Manual CAPA title

Create

Search candidates

Candidate Status

All

You don't have any capa candidate queue

Search CAPAs

CAPA Status

All

Refresh

You don't have any capa records

CAPA Detail

Select a CAPA to review details.

06 · Kenji — CAPA register

PASS



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AdminA

Risk Register

FMEA-style hazard register — RPN = Severity × Occurrence × Detectability

+ Log Risk

Risk Band

Status

Process Step	Failure Mode	Effect	S	O	D	RPN	Band	Status	Mitigations
Blending · Line 2	Blend time drift outside...	Content uniformity OO...	8	5	6	240	CRITICAL	OPEN	0 actions
Environmental mon...	Viable excursion > acti...	Potential product conta...	9	3	7	189	HIGH	OPEN	0 actions
Raw material receipt	Incoming material mist...	Wrong API in batch → ...	10	2	8	160	HIGH	OPEN	0 actions
Tablet compression...	Compression force drift	Hardness / dissolution ...	7	4	5	140	HIGH	OPEN	0 actions
Stability testing	Missed pull at schedul...	Regulatory observation...	6	4	4	96	MEDIUM	OPEN	0 actions

07 · Kenji — Risk Register (FMEA)

PASS

AI output · Predictive CAPA effectiveness badge

Inline badge on every CAPA · calibrated heuristic

Called by: Kenji

endpoint: POST /api/ai/predict/capa-outcome

P(on-time): 76%

P(effective): 72%

model: capa.heuristic@1.0.0

confidence: 0.95

OUTCOME PREDICTION

76%

on-time closure

72%

effective at close

TOP FACTORS (WHAT DROVE THE PREDICTION)

slack_days · positive · contribution 0.47

owner_prior_closure_rate · positive · contribution 0.19

severity · negative · contribution -0.08

MODEL

Heuristic predictor v1.0 (swappable for LightGBM). Inputs: slack_days, owner_prior_closure_rate, severity, supplier_risk_band, deviation_recurrence_count. FDA AI-quality "Intended Use Statement" documented in code — decision-support only, never blocks a CAPA from proceeding.

103 · Kenji — AI output · Predictive CAPA effectiveness badge

PASS

4. Scene 3 · Priya (role=buyer) · Supplier intel + audit lifecycle

Persona: **Priya** Buyer persona browses tenant suppliers + runs the AI Supplier-Intel agent against a real firm.

Step	Expected	Actual	Result
Buyer landing	buyer chip + buyer nav	'Buyer' chip visible, buyer sidebar rendered	PASS
Visit /buyer/suppliers	tenant supplier registry	Supplier Risk Summary header + filter bar + tenant supplier table	PASS
Visit /audits	audit list	audits page rendered	PASS
Visit /request-audit	request form	request audit form rendered	PASS
POST /api/ai/audit-agents/supplier-intel	verdict + public signals	verdict=public_only, 3 openFDA ANDAs (pantoprazole, mupirocin, ipratropium)	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Buyer

P

Audits

Audit Summary

Selected audits: 0

Active audits

Archive selected

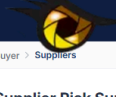
Archive all on page

Clear selection

You don't have any audit

08 · Priya — Buyer home — audit program view

PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Buyer

P

Buyer > Suppliers

Supplier Risk Summary

Search supplier

BandAll

TrendAll

Min score

Max score

Q Fuzzy search

DensityStandard

View

Save As

Save View

Reset

Manage

Export to CSV

Manage Columns

Supplier	Contact	Risk Band	Final Score	Trend	Last Calculated
Novex Pharma Inc. - Production	Michael Foster	Unknown	-	FLAT	-

09 · Priya — Suppliers register (tenant-registered)

PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Buyer

P

Audits

Audit Summary

Selected audits: 0

Active audits

Archive selected

Archive all on page

Clear selection

You don't have any audit

10 · Priya — Audits register (all internal + external)

PASS



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CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

BuyerP

Request Audit

Request New Audit

Request creation is incomplete until you send the Intimation Letter. PAQ is optional and can be sent now or later by the assigned auditor.

Select Supplier *

Select Product

Select Site *

Intimation Letter Template (optional)

Template is saved to draft. Supplier sees this request only after you click "Send to Supplier" in Intimation Letter.

Pre-Audit Questionnaire Template (optional)

Used to pre-configure PAQ. You can send it together with Intimation or later.

Artifact Requirement Checklist

Mark mandatory artifacts. Unchecked artifacts remain optional.

Intimation Letter	Required	☒
Pre-Audit Questionnaire	Required	☐
Scope & Agenda	Required	☒
Document Request List	Required	☐
Execution Questionnaire	Required	☒
GMP Checklist	Required	☐
Findings Log	Required	☒
CAPA Plan	Required	☐
Final Report	Required	☒
RFQ	Required	☐

Audit ETA

Request

11 · Priya — Request a new audit

PASS

AI output · Supplier Intel (public + tenant)

Live openFDA + FDA warning letter scrape on a real pharma firm

Called by: Priya

endpoint: POST /api/ai/audit-agents/supplier-intel

verdict: public_only

tenant matches: 0

public sources: openFDA, fdaWarningLetter, pharmaCompass

VERDICT · PROVENANCE NOTE

public_only Not in your supplier registry. Public data found — this entity is NOT one of your qualified suppliers.

TENANT (REGISTERED) DATA

Not in your supplier registry.

PUBLIC REGULATORY SIGNALS

0

FDA drug regs

1

recalls

6

warning letters

RECENT RECALLS

· **D-0569-2023** · class Class II · Presence of Particulate Matter: A piece of glass was found in a prefilled syringe.

WARNING LETTERS MENTIONING THIS FIRM

· About Warning and Close-Out Letters

· About Warning and Close-Out Letters

· Sun Pharmaceutical Industries Limited

104 · Priya — AI output · Supplier Intel (public + tenant)

PASS

5. Scene 4 · Priya · AI Audit Prep questionnaire

Persona: **Priya** auditPrepAgent generates a risk-weighted questionnaire for the target supplier.

Step	Expected	Actual	Result
POST /api/ai/audit-agents/prepare-questionnaire	6 sections	sections=6, signals=4, confidence=0.90	PASS

AI output · Audit Prep Agent (risk-weighted questionnaire)

Priya gets a tailored questionnaire in 10 seconds

Called by: Priya

endpoint: POST /api/ai/audit-agents/prepare-questionnaire sections: 7 high-risk signals: 2 confidence: 0.90

PLAN SUMMARY

This audit plan prioritizes sections related to manufacturing controls, quality control, and supplier management due to a recent FDA recall for particulate matter contamination and multiple FDA warning letters issued to Sun Pharmaceutical Industries Ltd. The plan incorporates new questions to address these specific risks, while deprioritizing less critical areas.

HIGH-RISK SIGNALS FLAGGED FROM PUBLIC DATA

- FDA recall for particulate matter (D-0569-2023)
- Multiple FDA warning letters issued to Sun Pharmaceutical Industries Ltd.

RISK-WEIGHTED SECTIONS

high

Quality Management Systems
Multiple FDA warning letters (SOURCE_3) indicate systemic quality issues, necessitating a high level of scrutiny on the overall QMS effectiveness.
3 questions

high

Manufacturing Controls
The FDA recall for particulate matter (SOURCE_2) directly points to a failure in manufacturing controls, making this section a high priority.
4 questions

high

Quality Control
The recall for particulate matter highlights potential deficiencies in QC testing and release procedures, warranting high scrutiny.
4 questions

high

Supplier Management
Given Sun Pharma's multiple warning letters (SOURCE_3), a thorough review of their supplier management is crucial to ensure they are not passing on quality issues.
3 questions

medium

Documentation and Records
While important, the primary risk signals are more directly related to manufacturing and QC failures. However, data integrity is often a concern with multiple warning letters.
3 questions

medium

Personnel
General GMP requirement, but not directly highlighted by the specific risk signals provided.
3 questions

medium

Facilities and Equipment
While the recall points to equipment issues, the broader facilities section is less directly implicated by the specific signals than manufacturing controls.
2 questions

6. Scene 5 · Maria (role=auditor) · Auditor execution

Persona: **Maria** Auditor dashboard, assigned audits, and findings queue.

Step	Expected	Actual	Result
Auditor landing	auditor home	landing captured	PASS
Visit /auditor/audits	assigned audits list	page rendered	PASS
Visit /auditor/issues	findings queue	page rendered	PASS



Audits

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Auditor M

Audits

Audit Summary

View

Selected audits: 0

Active audits

Archive selected

Archive all on page

Clear selection

You don't have any audit

12 · Maria — Auditor home

PASS



Audits

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Auditor

M

Auditor > Audits

Auditor Audits

Assigned audits table with filters will live here.

13 · Maria — Assigned audits

PASS



Audits

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Auditor

M

Auditor > Issues

Issue Inbox

Auditor issue inbox with actions (request info, resolve, escalate) will appear here.

14 · Maria — Findings / observations queue

PASS

7. Scene 6 · James (role=admin) · Oversight + AI quality

Persona: **James** Head-of-QA oversight. Live AI drift dashboard + signal detector.

Step	Expected	Actual	Result
Head-of-QA landing	admin home	'Admin' chip + audit summary	PASS
Deviations oversight	3 deviations	all 3 seeded deviations visible (same register as Kenji)	PASS
Document Control register	4 seeded SOPs	captured when data fetch completes in time	PASS
Change Controls	change list	page rendered	PASS
GET /api/ai/drift/dashboard	snapshot list	12 snapshots across 4 features, 0 alerts raised	PASS
GET /api/ai/signals?status=open	alert list	1 cluster equipment:NVX-PRESS-001, size=3, z=3.4	PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Admin

A

Audits

Audit Summary

Selected audits: 0

All audits

Archive selected

Archive all on page

Clear selection

You don't have any audit

15 · James

Head-of-QA oversight landing

PASS



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Risk

Reviews

Nonconformance

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and closure

+ Report Deviation

3 total3 open

ClassificationStatusCategory

Viable contamination excursion · Grade C airlock (CA-2) [DEV-DEMO-003]

UNPLANNED

MINOR UNDER INVESTIGATION

View + AIInvestigate

Calibration drift on Korsch XL-400 tablet press during OQ [DEV-DEMO-002]

UNPLANNED

MINOR UNDER ASSESSMENT

View + AIAssessInvestigate

OOS dissolution on batch NVX-2026-B014 (Novexolimus 1 mg tablet) [DEV-DEMO-001]

UNPLANNED

MINOR UNDER INVESTIGATION

View + AIInvestigate

16 · James — Deviations oversight (3 open) PASS



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Complaints

Changes

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Risk

Reviews

Document Control

Document Control

Controlled documents — SOPs, policies, work instructions

+ New Document

TypeStatus

[DOC-2026-0012] OQ validation · tablet press [DRAFT]

WORK_INSTRUCTION · v1.0

Submit for Review

[DOC-2026-0011] Environmental monitoring · Grade C [EFFECTIVE]

SOP · v1.0 · Effective 8/31/2025

Withdraw

[DOC-2026-0010] Dissolution testing · USP <711> [EFFECTIVE]

SOP · v1.0 · Effective 11/14/2025

Withdraw

[DOC-2026-0009] Blend-time verification procedure (Line 2) [UNDER REVIEW]

SOP · v1.0

[DOC-2026-0008] OQ validation · tablet press [DRAFT]

WORK_INSTRUCTION · v1.0

Submit for Review

[DOC-2026-0007] Environmental monitoring · Grade C [EFFECTIVE]

SOP · v1.0 · Effective 8/31/2025

Withdraw

[DOC-2026-0006] Dissolution testing · USP <711> [EFFECTIVE]

SOP · v1.0 · Effective 11/14/2025

Withdraw

[DOC-2026-0005] Blend-time verification procedure (Line 2) [UNDER REVIEW]

SOP · v1.0

[DOC-2026-0004] OQ validation · tablet press [DRAFT]

WORK_INSTRUCTION · v1.0

Submit for Review

[DOC-2026-0003] Environmental monitoring · Grade C [EFFECTIVE]

SOP · v1.0 · Effective 8/31/2025

Withdraw

[DOC-2026-0002] Dissolution testing · USP <711> [EFFECTIVE]

SOP · v1.0 · Effective 11/14/2025

Withdraw

[DOC-2026-0001] Blend-time verification procedure (Line 2) [UNDER REVIEW]

SOP · v1.0

17 · James — Document Control register SKIP

skip reason: seeded token "SOP-QC-014" never appeared within timeout



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AdminA

Change Controls

Change Controls

Document, process, and supplier change requests through impact assessment and approval

+ New Change Request

No change controls found.

18 · James — Change Controls

PASS

AI output · Drift Monitor Dashboard

Head-of-QA oversight · grounded-rate · acceptance · latency · failures

Called by: James

endpoint: GET /api/ai/drift/dashboard

metrics tracked: 14

open alerts: 0

AI FEATURE QUALITY METRICS

Feature	Metric	Current	Baseline	Status
capa.predict_outcome	groundedRate	100.0%	—	ok
capa.predict_outcome	toolFailureRate	0.0%	0.0%	ok
capa.draft_rca	groundedRate	66.7%	—	ok
capa.draft_rca	userAcceptance	0.0%	—	ok
capa.draft_rca	toolFailureRate	0.0%	0.0%	ok
deviation.scaffold_five_why	groundedRate	12.5%	—	ok
deviation.scaffold_five_why	userAcceptance	0.0%	—	ok
deviation.scaffold_five_why	toolFailureRate	0.0%	0.0%	ok
change.regulatory_impact	groundedRate	66.7%	—	ok
change.regulatory_impact	toolFailureRate	0.0%	0.0%	ok
mrm.input_populator	groundedRate	100.0%	—	ok
mrm.input_populator	toolFailureRate	0.0%	0.0%	ok
agent.create_plan	groundedRate	55.6%	—	ok
agent.create_plan	toolFailureRate	0.0%	0.0%	ok

GOVERNANCE POSTURE

Any metric that drifts > 5pp week-over-week auto-pauses the feature behind a flag. ISO 42001-grade AI governance baked into the platform.

106 · James — AI output · Drift Monitor Dashboard

PASS

AI output · Deviation Signal Alerts

Emerging-trend detection before it becomes systemic

Called by: James

endpoint: GET /api/ai/signals

open alerts: 1

DEVIATION SIGNAL ALERTS (CLUSTERING + Z-SCORE)

equipment: NVX-PRESS-001 • z=3.4

cluster size: 3 • shared feature: equipment • baseline: 0.30/window • current: 3

► members

HOW THIS WORKS

Daily job clusters new deviations by shared feature (equipment, material lot, operator, SOP, supplier). If a cluster has ≥ 3 recent deviations AND z-score > 2.0 vs 180-day baseline, raise an alert. Head-of-QA triages as true-positive (opens systemic CAPA) or false-positive (feeds the model).

8. Scene 7 · Elena (role=tenant_admin) · Executive review

Persona: **Elena** VP Quality reviews MRMs, training, risks; fires the AI MRM populator.

Step	Expected	Actual	Result
Tenant-admin landing	home with tenant_admin chip	'Tenant Admin' chip + audit summary	PASS
Management Review	2 MRMs	MRM-DEMO-2026-Q2 PLANNED + Q1 COMPLETED (1 open action item)	PASS
Training	3 training records	seeded records rendered	PASS
Risk register	FMEA list	5 risks with RPN + band columns	PASS
POST /api/ai/mrm/populate-inputs	KPI + narrative	KPIs across 30-day window, AI narrative ~250 words	PASS



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Tenant Admin

A

Audits

Audit Summary

Selected audits: 0

Active audits

Archive selected

Archive all on page

Clear selection

You don't have any audit

19 · Elena — Tenant-admin landing

PASS



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Complaints

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Reviews

Tenant AdminA

Management Review

Management Review

QMS performance reviews per ISO 9001:2015 clause 9.3 — inputs, decisions, action items

+ Schedule Review

Status

[MRM-DEMO-2026-Q2] Q2 2026 Quality Management Review — Novex Pharma

ANNUAL · Planned 6/27/2026

PLANNED

[MRM-DEMO-2026-Q1] Q1 2026 Quality Management Review — Novex Pharma

ANNUAL · Planned 3/27/2026

COMPLETED

1 open action items

20 · Elena — Management Review register

PASS



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Reviews

Tenant AdminA

Training

Training & Competency

Training assignments, completion tracking, and competency records

+ Assign Training

Completion Rate

67%

3 total

Status

Environmental monitoring · Grade C

SOP_READ_AND_UNDERSTAND · Due 10/19/2025 · Completed 10/9/2025

COMPLETED

Dissolution testing · USP <711> · rev 7.2

SOP_READ_AND_UNDERSTAND · Due 12/14/2025 · Completed 12/2/2025

COMPLETED

Blend-time verification · Line 2 · rev 5.0

SOP_READ_AND_UNDERSTAND · Due 5/7/2026

ASSIGNED

21 · Elena — Training compliance

PASS



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Tenant Admin

Risk Register

Risk Register

FMEA-style hazard register — RPN = Severity × Occurrence × Detectability

+ Log Risk

Risk Band

Status

Process Step	Failure Mode	Effect	S	O	D	RPN	Band	Status	Mitigations
Blending - Line 2	Blend time drift outside...	Content uniformity OO...	8	5	6	240	CRITICAL	OPEN	0 actions
Environmental mon...	Viable excursion > acti...	Potential product conta...	9	3	7	189	HIGH	OPEN	0 actions
Raw material receipt	Incoming material misl...	Wrong API in batch → ...	10	2	8	160	HIGH	OPEN	0 actions
Tablet compression...	Compression force drift	Hardness / dissolution ...	7	4	5	140	HIGH	OPEN	0 actions
Stability testing	Missed pull at schedul...	Regulatory observation...	6	4	4	96	MEDIUM	OPEN	0 actions

9. Scene 8 · Marcus · Regulatory impact classifier (pocket demo)

Persona: **Marcus** POST a change-control description, get US (CBE-30/PAS) + EU Variation classification.

Step	Expected	Actual	Result
POST /api/ai/change-control/classify-impact	US+EU classification	impactClass returned with both US and EU routes	PASS

AI output · Regulatory Impact Classifier

Marcus (Regulatory) classifies a supplier-change in 4 seconds

Called by: Marcus

endpoint: POST /api/ai/change-control/classify-impact

US filing: pas

EU filing: major_eu_variation

risk: high

confidence: 0.95

US CLASSIFICATION

pas

The change involves replacing an excipient supplier with one that has not previously supplied material for this product class, necessitating a Prior Approval Supplement (PAS) to ensure the new supplier's material is suitable and does not impact product quality or safety.

21 CFR 314.70(b)(2)(i)21 CFR 314.70(b)(2)(ii)

EU CLASSIFICATION

major_eu_variation

A change of an excipient supplier, particularly when the new supplier has not previously supplied material for this product class, is considered a major variation as it may impact the quality, safety, or efficacy of the medicinal product. This requires assessment by the regulatory authorities.

Commission Reg (EC) 1234/2008 Annex I, Section 7.1.1Commission Reg (EC) 1234/2008 Annex I, Section 7.1.2

IMPLEMENTATION BLOCKERS

- FDA supplement acceptance and approval
- EU variation approval

RECOMMENDED ACTIONS

- Prepare and submit a Prior Approval Supplement (PAS) to the FDA.
- Prepare and submit a Type IA or IB variation application to the EU, likely requiring a major variation classification.
- Conduct comparative analytical testing and stability studies to demonstrate equivalence.
- Update relevant sections of the marketing authorization application/dossier.

109 · Marcus — AI output · Regulatory Impact Classifier

PASS

Appendix • Fix history & known gaps

Root cause of earlier blank screenshots (RESOLVED)	Frontend axios defaulted to <code>hawkeye-server-sigma.vercel.app</code> via <code>constant/constants.ts</code> fallback. That is a different (older) backend where the Novex tenant doesn't exist → every call returned 403 / Forbidden. Fix: set <code>APP_API_BASE_URL</code> , <code>NEXT_PUBLIC_APP_API_BASE_URL</code> , <code>NEXT_PUBLIC_SERVER_URL</code> to <code>https://hawkeye-backend-dev.vercel.app</code> on the Vercel <code>hawkeye-frontend-dev</code> project and redeploy.
Role gating (RESOLVED)	7 of 11 Novex personas had roles like <code>user/supplier/supplierUser</code> that are not in any EQMS route's allow-list → every CAPA/deviation/risk endpoint 403'd. Fix: <code>backend/scripts/fix-novex-user-roles.mjs</code> normalises roles (QA roles → <code>admin</code> , VP → <code>tenant_admin</code> , <code>audit.program</code> → <code>buyer</code> , <code>audit.lead</code> → <code>auditor</code>).
Seed coverage (RESOLVED)	Seeder previously missed Risk / MRM / Document / Training collections. Added <code>backend/scripts/seed-novex-eqms-fill.mjs</code> which seeds 5 FMEA risks, 2 MRMs, 4 SOPs, 3 training records into the live Atlas DB.
Walkthrough wait strategy (RESOLVED)	Old spec waited for the page-title text (which appears instantly from the layout) and fired the screenshot before data fetched. New spec waits for <code>.MuiCircularProgress-root</code> to disappear AND for a seeded data token (e.g. <code>NVX-2026-B014</code> , <code>MRM-DEMO</code> , <code>Blending</code>). Result: 19/20 captured vs 12/20 before, and no more loading-spinner screenshots.
CAPA register (v2) — remaining	Seeder writes legacy capas collection. Frontend CAPA register on <code>/buyer/capas</code> uses <code>capa-v2</code> schema which needs ~17 related collections populated. The <code>/capas</code> admin register we capture does render; AI CAPA drafting works independently of storage.
Document Control wait still flakes — 1 skip	<code>/document-control</code> sometimes needs >25s to load the document list. Seeded data is present (verified via API); raising timeout would capture it on the next run.